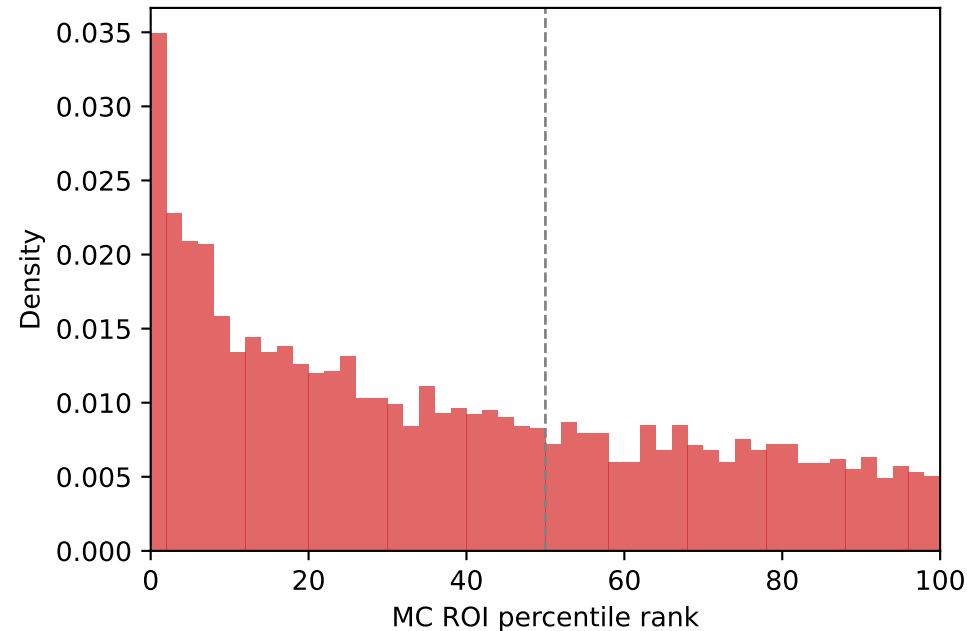
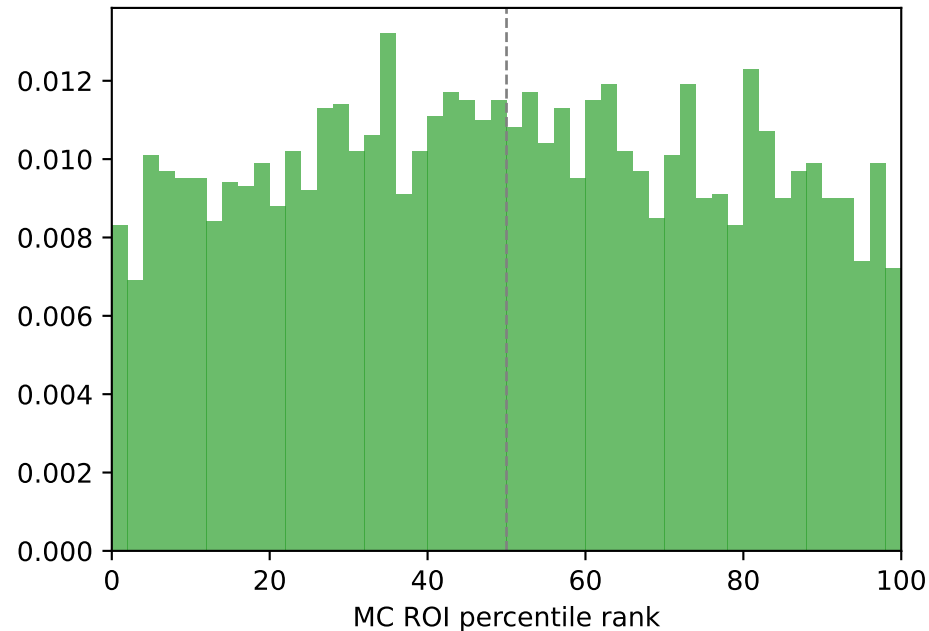


Floating-point summation order produces a spurious leftward shift in the MC ROI rank distribution (iid Normal pnl's; permutation null trivially exchangeable)

(a) Repo procedure as written
 $\texttt{` (obs\_roi > sh\_roi).sum() / B * 100 `}$
mean=37.6 std=29.1 % < 50 = 66.6
(matches paper Table 4: 40.8 / 29.2 / 62.6)



(b) Repaired: mid-rank tie handling
 $\texttt{` (lt + 0.5*eq) / B * 100 `}$
mean=49.9 std=27.7 % < 50 = 50.4
(centred at 50, correct for permutation-invariant stat)



(c) Repaired: precise sorted summation
 $\texttt{` precise\_sum(pnl's) `}$ (no FP order ambiguity)
mean=0.0 std=0.0 % < 50 = 4.0
(rank collapses to 0 — the correct degenerate result)

